| S.No: 12 | Exp. Name: ***Longest Common Sub Sequence*** | Date:2026-10-07 |
|---|---|---|





**Aim:**

Write a C program that takes three strings $X$, $Y$, and $Z$ as input and finds the length of the Longest Common Subsequence (LCS) that is present in all three strings.

**Input Format:**

- The first line contains the string $X$.
- The second line contains the string $Y$.
- The third line contains the string $Z$.

**Output Format:**

- A single integer representing the length of the Longest Common Subsequence among the three given strings.

**Constraints:**

- $1 \leq length\ of\ X \leq 10^2$
- $1 \leq length\ of\ Y \leq 10^2$
- $1 \leq length\ of\ Z \leq 10^2$

**Sample Test Case:**

**Input:**

```
Twenty
Thirty
Forty
```

**Output:**

```
2
```

**Explanation:**

- The Longest common sub-sequence in the given three strings is "**ty**" so the answer is 2 because it is having length of 2.

**Source Code:**

lcss.c

```
#include<stdio.h>
#include<string.h>
#include<stdbool.h>
#define MAX_LENGTH 100

int max(int a, int b){if(a>b){return a;}else{return b;}}

int lcsOf3(char X[], char Y[], char Z[], int m, int n, int o){

        //write your code here...
if (m > 0 && X[m - 1] == '\n') X[--m] = '\0';
    if (n > 0 && Y[n - 1] == '\n') Y[--n] = '\0';
    if (o > 0 && Z[o - 1] == '\n') Z[--o] = '\0';

    // 3D DP Array to track common subsequences
```

```
    int dp[m + 1][n + 1][o + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            for (int k = 0; k <= o; k++) {
                if (i == 0 || j == 0 || k == 0) {
                    dp[i][j][k] = 0;
                } else if (X[i - 1] == Y[j - 1] && X[i - 1] == Z[k - 1]) {
                    dp[i][j][k] = dp[i - 1][j - 1][k - 1] + 1;
                } else {
                    // Fix: Use the provided max function sequentially
                    dp[i][j][k] = max(max(dp[i - 1][j][k], dp[i][j - 1][k]), dp[i][j]
[k - 1]);
                }
            }
        }
    }

    return dp[m][n][o];







}

int main()
{       char x[MAX_LENGTH], y[MAX_LENGTH],z[MAX_LENGTH];
    fgets(x, MAX_LENGTH, stdin);
    fgets(y, MAX_LENGTH, stdin);
    fgets(z, MAX_LENGTH, stdin);
        printf("%d", lcsOf3(x, y, z, strlen(x), strlen(y), strlen(z)));

        return 0;
}
```

## Execution Results - All test cases have succeeded!

| Test Case - 1 |
|---|
| User Output |
| Twenty |
| Thirty |
| Forty |
| 2 |

| Test Case - 2 |
|---|
| User Output |
| One |
| Two |
| Ten |
| 0 |

| Test Case - 3 |
| --- |
| User Output |
| coder |
| coding |
| codetantra |
| 3 |